

## Lecture 12

### Attendance Questions

- Continuous surjections between compacts are homeos

### Peano Curves

- A Peano space  $M$  is a space that's a continuous image of  $[0, 1]$ 
  - Compact because  $[0, 1]$  is
  - Not necessarily locally path connected but always path connected
    - Think about unit interval with vertical line segments at the end points and at  $\frac{1}{n}$
    - No local path between end of vertical segment at 0 and some small  $\frac{1}{n}$
- A space is locally path connected if  $\forall \varepsilon > 0, \exists \delta$  such that  $d(x, y) < \delta \Rightarrow \text{path in } M_\varepsilon(x)$

### Cantor Surjection Theorem

- Hausdorff & Alexandrov
- Every compact metric space is the continuous image of a Cantor set  $C$

### Peano Curve for $D^2 \subset \mathbb{R}^2$

- Since  $D^2$  is closed and bounded in  $\mathbb{R}^2$ , it is compact
- There is a continuous surjection from  $C \rightarrow D^2$
- Extend  $f : C \rightarrow D^2$  to  $\hat{f} : [0, 1] \rightarrow D^2$
- For gap intervals in  $C$ , we parametrize
  - $x = (1 - t)a + tb$
  - $\hat{f}(x) = (1 - t)f(a) + tf(b)$
- Can prove without Cantor Surjection
  - Split a square into smaller squares, form paths, then shrink
  - Lots of  $\varepsilon$ , Cauchy sequence of paths
  - Produces Peano curve

### Lakes of Wada

- Answers the question: what is a curve?
- Island nation, lake of freshwater, seawater around
- Day 1: dig canal connecting to sea, within 1 mile of every point in the island
- Day 2: same procedure but connecting to lake, no overlap
- Repeat but shorten distances
- Pile of sand principle
- Canals curve most of space
  - Is shoreline a curve?
- Think of 2 lakes now, 1 hot, 1 cold
  - Is the common boundary a curve?
  - 3 open sets w/ common boundary

### Proof of Cantor Surjection

- Somewhat similar process to Moore-Klein
  - Thinking of breaking down into specific points for Moore Klein
  - Thinking of nested sequences for Cantor Surjection

**Lemma:** If  $M$  is compact,  $M$  can be expressed as a finite number of  $\varepsilon$ -bits.  $\text{diam}_{\frac{\varepsilon}{2}}\text{-bit is } \leq \varepsilon$

- Make a covering by neighborhoods  $\{M_{\frac{\varepsilon}{2}}(p), p \in M\}$
- Reduces to a finite subcovering that covers  $M$

## Labeling with Words

- $\varepsilon_n \downarrow 0$
- $\exists$  finite covering by  $\varepsilon_1$ -bits of  $M$
- $\mathcal{M}_1 = \mathcal{M}(M)$
- $\mathcal{M}_2$  refines  $\mathcal{M}_1$ 
  - $K \cap \mathcal{M}_2$  is some collection of  $\varepsilon_2$ -bits
- For  $c \in C$ ,  $f(c) =$  a point in  $M$ 
  - $c = (c_1 c_2 \dots)$  base 3
- List elements of covering using words
  - $\sigma_1 = \omega(k_1) \rightarrow \mathcal{M}_1$  a surjection
    - Just choose some big  $k_1$ ,  $\mathcal{M}_1$  contains a finite number of elements
- Looking at more words using  $\sigma_2 : \omega(k_1) \times \omega(k_2) \rightarrow \mathcal{M}_2$
- Continue refining
- Labeling for  $\mathcal{M}_i$  is not one-to-one, but is onto
- $c \in C$ ,  $c = (c_1 c_2 \dots)$ 
  - Prefix of length  $k_1$  corresponds to a word in  $\omega(k_1)$
  - Continue on and on, create nested sequence of subsets

## Continuity

- 2 members close in  $C$ , then word label shares prefix
- Corresponds to same nested sequence of sets
- $\sigma$  is some small set at the end, so the values are close in  $M$

## Surjection

- $p \in M$  is in some  $K \subset \mathcal{M}_1$
- Some  $\varepsilon_2$ -bits contains  $p$
- Corresponds to some word, meaning it corresponds to some  $c \in C$